EBN111 Class Test / *Klastoets* 1 (3 Mar 2017)

15 Marks / *Punte* 30 Minutes / *Minute*

Question 1 [4]

a) The lamp beside your bed uses an old 100 W tungsten globe. You replace the old globe with a 5 W LED globe. How much money do you save over one year (365 days) if the light is on for 90 minutes per day and electricity costs R1.6 per kWh?

b) The energy associated with a certain element in a circuit decreases linearly from 20 *mJ* to 10 *mJ* between 0 *ms* and 4 *ms*. If a constant voltage of 10V is applied over the element, what is the current through the element after 2 *ms*? (Assume current flows from the positive to the negative voltage terminal through the element.)

Vraag 1 [4]

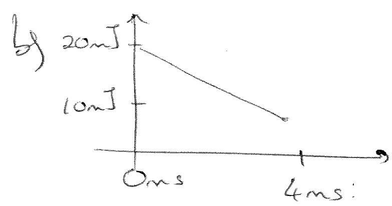
a) Die lamp langs jou bed gebruik 'n ou 100 W tungsten gloeilamp. Jy vervang die ou gloeilamp met 'n 5 W LED gloeilamp. Hoeveel geld bespaar jy oor een jaar (365 dae) indien die gloeilamp aan is vir 90 minute per dag en elektrisiteit kos R1.6 per kWh?

b) Die energie geassosieer met 'n sekere element in 'n kringbaan val linêer van 20 *mJ* tot 10 *mJ* tussen 0 *ms* en 4 *ms*. Indien daar 'n konstante spanning van 10 V oor die element is, wat is die stroom wat deur die element vloei na 2 *ms*? (Aanvaar die stroom vloei vanaf die positiewe na die negatiewe spanningsterminaal deur die element.)

a) $R = 1.6\ R/kWh \times 1.5h/day \times 365\ days \times (0.1 - 0.005)kW$

$= R83.22$

b)



$p = \frac{dw}{dt} = v \cdot i$

At $t = 2ms$:

$\frac{-10mJ}{4ms} = 10 \cdot i \Rightarrow i = -250\ mA$

Question 2 [4]

If $V_0 = 140$ V, calculate the power associated with the

a) 50 V voltage source (P_V), and

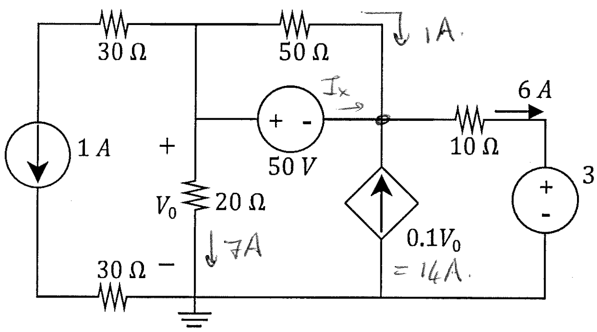
b) the dependant current source (P_A).

Vraag 2 [4]

Indien $V_0 = 140$ V, bepaal die drywing geassiosieer met

a) die 50 V stroombron (P_V), en

b) die afhanklike stroombron (P_A).

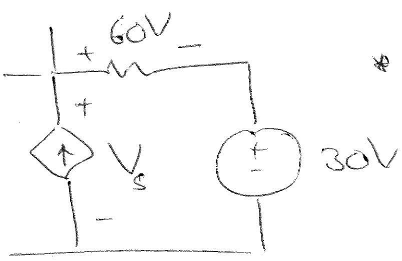


* $I_x = 6 - 1 - 14$

$= -9A$

* $P_V = +50 \cdot I_x$

$= -450W$



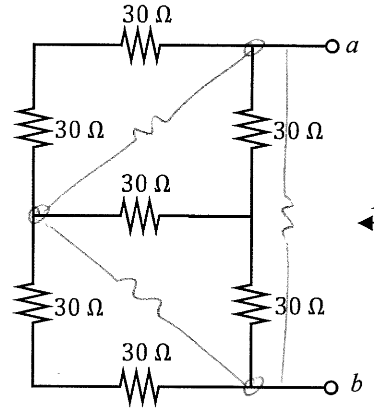
* $-V_s + 60 + 30 = 0$

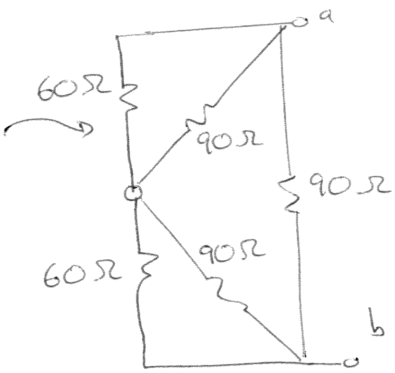
$V_s = 90V$

* $P_A = -(0.1V_0)(90)$

$= -1.26kW$

Question 3 [2]	Vraag 3 [2]
Determine R_{ab}.	Bepaal R_{ab}.





$R_\Delta = 3R_Y$

$R_{ab} = (90\|60 + 90\|60)\|90$

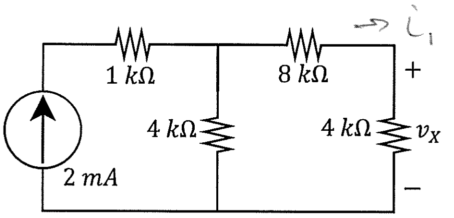
$= (36 + 36)\|90$

$= 72\|90$

$= \frac{72 \times 90}{72+90}$

$= \underline{40\,\Omega}$

Question 4 [2]	Vraag 4 [2]
Determine v_X.	Bepaal v_X.



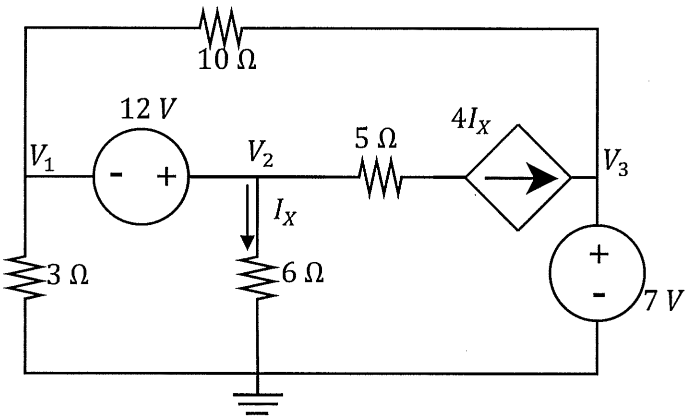
* $i_1 = \frac{2mA \times 4k}{4k+8k+4k}$ (I-division).

$= 0.5\,mA.$

* $v_x = +0.5m \times 4k.$

$= \underline{2V}$

Question 5 [4]	Vraag 5 [4]
Use nodal analysis to set up two equations of the following form: $aV_1 + bV_2 = 12$, $cV_1 + dV_2 = 21$	Gebruik nodeanalise om twee vergelykings van die volgende vorm op te stel: $aV_1 + bV_2 = 12$, $cV_1 + dV_2 = 21$



* $V_2 - V_1 = 12V$

$\therefore \underline{-V_1 + V_2 = 12V}$

* $I_x = +\frac{V_2}{6}$

* $\frac{V_1}{3} + \frac{V_1 - 7}{10} + 5I_x = 0.$

$\therefore \frac{V_1}{3} + \frac{V_1}{10} + 5\cdot\frac{V_2}{6} - \frac{7}{10} = 0$

(×30) $10V_1 + 3V_1 + 25V_2 - 21 = 0$

$\underline{13V_1 + 25V_2 = 21}$